

Michail Tamiolakis

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SUMMARY

Software Engineer specializing in Unity, Physics, C++, C#, and Multiplayer XR applications, with professional experience developing VR medical simulation software and leading engineering teams. MSc in Computer Science and Engineering with research experience in high-performance soft-body simulation, cutting, tearing, and piercing interactions for standalone and mobile XR applications.

EXPERIENCE

ORamaVR

Sep. 2024 – Jun. 2025

Lead SDK Developer

Heraklion, Crete

- Led a development team of 2-5 engineers, overseeing technical direction, task planning, daily coordination, and progress tracking.
- Designed and implemented a new physics engine for interactive soft-body simulations, delivering over 150% higher performance than the legacy solution and 39% higher performance than the state-of-the-art solution.
- Designed and implemented a hardware-agnostic and extensible input management framework, with integrations for OpenXR and the Meta SDK.
- Designed and implemented a hand interaction system with automated pose estimation, improving usability and reducing simulation development time by 20%.
- Conducted code reviews, provided technical support and guidance to developers, and onboarded new team members.
- Estimated feature development timelines, decomposed high-level requirements into development tasks, and created implementation tickets.
- Developed tools that reduced release and testing cycle times by approximately 15%.
- Developed internal tools to enhance team productivity and optimized CI/CD workflows to improve development efficiency.
- Designed and implemented a Unity authentication tool with support for Google Single Sign-On (SSO).

ORamaVR

Apr. 2022 – Aug. 2024

Experienced VR Developer

Heraklion, Crete

- Designed and implemented new SDK features for simulation-specific tasks based on application requirements.
- Used the Unity Test Framework to create tests for new SDK features.
- Created and maintained SDK documentation for new features using Sphinx.

ORamaVR

Jun. 2021 – Mar. 2022

Junior VR Developer

Heraklion, Crete

- Used the MAGES SDK to create interactive medical simulation mechanics.
- Performed QA testing and updated SDK documentation using Sphinx.

PROJECTS

Real-Time Physics Framework for Standalone VR Applications (demo reel available on request – code under NDA)

- Developed a real-time physics framework as part of my MSc research and subsequently extended it into a complete physics and interaction system for standalone VR applications.
- Designed and implemented a physics engine for Unity based on Extended Position-Based Dynamics (XPBD).
- Implemented a custom collision detection system using spatial hashing grids.
- Developed simulation support for soft bodies, rigid bodies, cloth, ropes, and stiff rods.
- Implemented real-time cutting and tearing interactions for soft bodies, ropes, and rods.
- Implemented real-time piercing interactions using custom constraint systems.

- Highly optimized the framework using a **parallel, data-oriented architecture**, leveraging the **Burst Compiler, Native Collections, and vectorized instructions**.
- Currently developing an open-source, ground-up reimplementation of the framework, as the original codebase is proprietary under NDA.

BT201 Arduino Library | [GitHub](#)

- Designed and implemented an open-source Arduino library for interfacing with the BT201 Bluetooth module over Serial.
- Tested the library in a real-world smart speaker application.

RESEARCH

MAGES 4.0: Accelerating the World's Transition to VR Training and Democratizing the Authoring of the Medical Metaverse

IEEE Computer Graphics and Applications

- Contributed to the development of the MAGES SDK, with a primary focus on physics systems and interactive mechanics for collaborative medical VR/AR training applications.

Progressive Tearing and Cutting of Soft-Bodies in High-Performance Virtual Reality

ICAT-EGVE 2022 – Hiyoshi, Yokohama, Japan

- Developed a highly optimized algorithm for real-time cuts and consecutive tears on deformable rigged and soft-body mesh models for standalone VR applications.

EDUCATION

University of Crete

Master of Science in Computer Science and Engineering

Sep. 2023 – Mar. 2025

Heraklion, Greece

- **MSc Thesis:** A Real-Time Physics Framework with Advanced Cutting, Tearing and Piercing Interactions for Mobile XR Applications
- **GPA:** 9.2/10

University of Crete

Bachelor of Science in Computer Science and Engineering

Sep. 2017 – Mar. 2022

Heraklion, Greece

- **BSc Thesis:** An Adaptive, Optimized Soft-Body Mesh Algorithm for Interactive Skinned Geometrical Meshes in VR
- **GPA:** 7.6/10

SKILLS

Languages: C#, C++, HLSL

Technologies: Unity, Unity DOTS, OpenXR, Meta SDK

Domains: VR, AR, XR, Real-Time Physics Simulation

Physics: XPBD, Collision Detection, Soft-Body Simulation, Real-Time Cutting, Tearing & Piercing

Systems: Networking, Parallel Programming, Data-Oriented Programming, CI/CD, Git

LANGUAGES

Greek (Native) English (C2, ECPE) German (B2, Goethe-Zertifikat)